AI

Everyone Is Talking About AI. No One Tells You What You Need to Learn

The exact AI skills to learn, in order, for data and tech careers. What to learn first, what to skip, and a 90-day plan.

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First, pick your level

Most people stall because they try to learn everything. AI skills come in 3 levels. Most jobs need level 1 and 2.

- **Level 1: Use AI well**

  Everyone. Get faster and better at your current work with AI tools.

- **Level 2: Work with AI as an analyst**

  Data and tech roles. Use AI for SQL, Python, analysis, and understand how it works.

- **Level 3: Build with AI**

  Developers and data scientists. Connect AI models to data and apps through code.

- **Not needed for most jobs**

  Training large models from scratch. That is research work at a small number of companies.

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Level 1: Use AI well

These 4 skills apply to every role, technical or not.

Write clear prompts

Give context, the task, the format you want, and an example. Vague prompts give vague answers.

Verify everything

AI can state wrong facts confidently. This is called hallucination. Check numbers, names and code before you use them.

Protect data

Never paste confidential customer, salary or company data into a public AI tool. Follow your company's AI policy.

Know where it is weak

Exact calculations on large data, very recent events, and anything it cannot see. Use AI for drafts, not final answers.

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A prompt that works

Use this structure for almost any task. Copy it.

```
ROLE:    You are a senior data analyst.
CONTEXT: I have a sales table with columns
         order_id, order_date, region, amount.
TASK:    Write a SQL query for month-on-month
         revenue growth % by region.
FORMAT:  PostgreSQL. Explain each step
         in one line.
CHECK:   List 2 ways this query could give
         a wrong result.
```

Why the last line matters

Asking AI to find weak points in its own answer catches many mistakes before you run anything.

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The foundations matter more now

AI makes skilled people faster. It does not replace the skill. You must be able to judge its output.

- **SQL**

  AI writes queries. You must spot a wrong join or a missing filter.

- **Python and Pandas**

  You must read and fix the code AI gives you.

- **Statistics**

  You must know if an AI-suggested conclusion is real or noise.

- **Business understanding**

  AI does not know your company, customers or goals. You do. This is your edge.

Rule: if you cannot check the answer, you cannot safely use the answer.

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Level 2: How AI works

Know these terms in plain words. They come up in interviews and at work.

- **LLM**

  Large language model. Predicts the next piece of text based on patterns learned from huge amounts of text.

- **Token**

  The unit of text a model reads and writes. Roughly part of a word. Usage and cost are counted in tokens.

- **Context window**

  The maximum amount of text the model can consider at once, including your prompt and its reply.

- **Temperature**

  A setting for randomness. Lower gives more consistent answers, higher gives more varied ones.

- **Hallucination**

  Confident but wrong output. The reason verification is a core skill.

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Level 2: Use AI in analysis work

The highest-value uses for analysts today.

- **Write and review SQL and Python**
  Draft code faster, then ask AI to review your own code for bugs and edge cases.
- **Explain unfamiliar code**
  Paste a long query and ask what each part does.
- **Clean and categorise text**
  Classify feedback, tickets or reviews into themes, then check a sample by hand.
- **Summarise results**
  Turn a table of results into a first-draft summary for managers, then verify every number.

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Level 3: Build with AI

Learn these in order. Each one builds on the last.

1. Call a model from Python

Use an AI provider's API: send a prompt, get a response in code.

2. Structured output

Ask for JSON with fixed fields, so the result can go straight into a table.

3. Embeddings

Text turned into numbers that capture meaning. Similar texts get similar numbers. Used for search.

4. RAG

Retrieval-augmented generation. Find relevant documents first, then give them to the model to answer from. Reduces made-up answers.

5. Agents and tool use

The model decides which tool or function to call, like running a query, then uses the result.

6. Evaluation

Test outputs against expected answers on many examples. This is what makes an AI project reliable.

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What to skip for now

These take months and rarely help in data analyst or junior roles.

- **Training LLMs from scratch**
  Needs huge data, compute and research teams.
- **Deep transformer maths first**
  Useful later for ML research roles. Not needed to use or build with AI.
- **Chasing every new tool**
  Tools change every few months. The skills above stay useful.
- **Collecting certificates**
  One working project shows more than five certificates.

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Your 90-day plan

One hour a day. Adjust the pace to your level.

- **Days 1-15**

  Level 1: use AI daily at work. Practise the prompt structure and verify every output.

- **Days 16-45**

  Strengthen SQL, Python and statistics with AI as your tutor, not your answer key.

- **Days 46-60**

  Level 2: learn the terms on page 6. Use AI to classify a real text dataset and check its accuracy on 50 rows.

- **Days 61-90**

  Level 3 basics: call a model from Python, get JSON output, build a small RAG search over your own documents.

Project to show: an analysis where AI did the first draft and you documented what you checked and fixed. That proves judgement, the skill companies want most.

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